


# Autonomous AI Multi-Agent Orchestration Platform

System Architecture, ML Task Routing, RAG Vector Search & Production Runbook

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|---|---|
| **Document Version:** | 1.0.0 (Production Verified) |
| **Status:** | 17/17 Pytest Verified \| 99.32% ML Router Accuracy |
| **Technology Stack:** | FastAPI, React 19, Vite, scikit-learn, MongoDB Atlas, Ollama, Gemini |
| **Repository:** | github.com/srivastavaayush084/Autonomous-AI-Multi-Agent-Orchestration-Platform- |
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**EXECUTIVE NOTICE:** This technical manual encapsulates the complete design, mathematical benchmarks, security gateways, API specifications, and operational directives of the Autonomous AI Multi-Agent Platform. All components have been audited for zero data leakage, strict origin whitelisting, and deterministic DAG state enforcement.

# 1. Executive Summary & Core Value Proposition

Modern enterprise workflows demand automated goal decomposition capable of orchestrating diverse specialist agents, managing fine-grained dependencies, upholding data sovereignty, and providing air-tight safety gates. The **Autonomous AI Multi-Agent Orchestration Platform** addresses these imperatives through a five-pillar foundation:

- **Stateful DAG Task Engine:** Decomposes high-level goals into topological execution graphs with verifiable acceptance criteria, concurrent branch execution, and automated checkpoint recovery.
- **Audited ML Model Router:** A hybrid classifier combining rule-based heuristics with a trained scikit-learn RandomForest model achieving 99.32% test accuracy and cutting cloud API expenditure by 87.8%.
- **Data Sovereignty Firewall:** Automated policy engine screening task prompts for confidential data and enforcing 100% on-premise execution via local Ollama models (Qwen 2.5, DeepSeek Coder, Llama 3.2).
- **Retrieval-Augmented Generation (RAG):** 768-dimensional vector search integrated with MongoDB Atlas Vector Search ($vectorSearch), SHA-256 chunk deduplication, and strict project tenant isolation.
- **Human-in-the-Loop Safety Gateway:** Consequential tasks (deploy, delete, write, charge) are quarantined in waiting_approval, completely immune to direct API bypass (HTTP 403 Forbidden).

# 2. Multi-Agent Topology & Lifecycle State Machine

The platform coordinates specialized autonomous agent personas, each governed by role-specific prompt contracts:

| Agent Role | Core Responsibility | Primary Model Fleet | Artifact Deliverable |
|---|---|---|---|
| **Planner Agent** | Goal decomposition, topological DAG generation, dependency resolution | qwen2.5:latest | Execution Graph JSON |
| **Researcher Agent** | Domain knowledge discovery, constraint gathering, documentation synthesis | llama3.2:latest / Gemini 3.5 | Research & Strategy Briefs |
| **Developer Agent** | Software construction, code refactoring, schema design, script generation | deepseek-coder:6.7b | Executable Source Code |
| **Analyst Agent** | Metric evaluation, numerical modeling, telemetry & data analysis | deepseek-r1:8b / Gemini 3.5 | Analytical Forecasts |
| **Reviewer Agent** | Verification against acceptance criteria, QA tests, safety gating | qwen2.5:latest | Verification Audit Logs |

### Finite State Machine Transitions:

Every task adheres to strict, deterministic state transitions: PENDING → READY → RUNNING → [WAITING_APPROVAL] → COMPLETED / FAILED. The orchestrator continuously monitors completed upstream task IDs to safely unlock downstream dependencies in parallel branches.

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# 3. Machine Learning Model Router & Cost Audit

To balance quality, inference speed, and budget, the platform implements a two-stage routing pipeline. A feature extractor maps incoming task descriptors into 5 numerical dimensions: (1) Specialist Role, (2) Prompt Word Count, (3) Complexity Estimate, (4) Safety Gate Flag, and (5) Code Keyword Signals. The classifier was evaluated under strict deduplication across 589 task samples (441 train / 148 test holdout):

| Metric Dimension | Score / Value | Evaluation Methodology |
|---|---|---|
| **Stratified 5-Fold Cross Validation** | **99.66% (±0.42%)** | 5-Fold Cross Validation on 441 train partition |
| **Held-Out Test Accuracy** | **99.32% (147 / 148)** | Evaluated against unseen deduplicated test set |
| **Held-Out Macro F1 Score** | **99.13%** | Macro-averaged across all 5 task classes |
| **Train/Test Data Leakage** | **0.00%** | Zero overlapping samples between partitions |
| **Rule Baseline Accuracy** | 71.62% | Heuristic capability-based baseline |
| **ML Performance Lift** | **+27.70% Lift** | Accuracy gain over heuristic baseline |
| **Local Execution Ratio** | **87.8% Local** | 130 tasks routed on-premise; 18 to cloud API |
| **Cloud Cost Reduction** | **87.8% Savings** | Calculated under Gemini 3.5 Flash pricing tier |

### Confusion Matrix on 148 Held-Out Test Samples:

| Actual \ Predicted | coding | creative | data_analysis | reasoning | simple |
|---|---|---|---|---|---|
| **coding** (30) | 30 | 0 | 0 | 0 | 0 |
| **creative** (25) | 0 | 25 | 0 | 0 | 0 |
| **data_analysis** (30) | 0 | 0 | 29 | 1 | 0 |
| **reasoning** (33) | 0 | 0 | 0 | 33 | 0 |
| **simple** (30) | 0 | 0 | 0 | 0 | 30 |

# 4. Data Sovereignty & Enterprise Egress Guard

To ensure compliance with corporate governance, GDPR, and HIPAA standards, the platform includes a strict **Data Sovereignty Guard** (_apply_data_sovereignty_guard). Tasks tagged with confidentiality constraints or internal IP indicators are automatically intercepted before cloud dispatch and forced into on-premise execution via local Ollama inference. Cloud APIs are strictly forbidden from receiving sovereign payloads.

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# 5. Retrieval-Augmented Generation & Project Memory

The RAG subsystem provides long-term project memory and factual grounding for all specialist agents. Key operational properties include:

- **Recursive Chunking:** Configurable chunk sizes (500 tokens) with 50-token sliding overlap to preserve contextual continuity across boundaries.
- **SHA-256 Deduplication:** Identical documents uploaded multiple times are recognized via content hashing; chunk generation is bypassed, eliminating duplicate vector explosion.
- **Atlas Vector Search ($vectorSearch):** Semantic retrieval powered by 768-dimensional embeddings and cosine similarity index (project_embeddings_vector_index).
- **Tenant Isolation:** Every vector query injects filter: {'project_id': project_id}, mathematically preventing cross-tenant information leakage.
- **Cascading Deletion:** Project deletion or individual document purge atomically cleans both MongoDB metadata documents and vector embedding chunks.

# 6. Human-in-the-Loop Safety Gateway & Verification

Autonomous operations involving external risk (e.g. database migration, cloud deployment, asset purging, or payment authorization) are protected by cryptographic and role-based safety gates:

- **Automated Consequential Detection:** Tasks containing critical keywords (deploy, drop, delete, terminate, wipe, charge) are unconditionally marked for human approval.
- **Anti-Bypass Guard:** Direct API calls attempting to transition a gated task directly to RUNNING or COMPLETED without approved review records are rejected with **HTTP 403 Forbidden**.
- **Self-Healing Execution:** Once an authorized operator approves the request, the orchestrator immediately sets the task to READY and automatically resumes execution across all downstream branches.

# 7. Comprehensive REST API Reference

| Method & Endpoint | Description | Request / Response Payload |
|---|---|---|
| POST /api/v1/projects | Create isolated workspace | { name, description } |
| DELETE /api/v1/projects/{id} | Cascading delete project & vectors | Returns deletion confirmation |
| POST /api/v1/runs | Launch autonomous execution run | { project_id, goal } |
| POST /api/v1/runs/{id}/resume | Checkpoint recovery for paused run | Relaunches background DAG loop |
| GET /api/v1/runs/{id}/tasks | List all tasks with self-healing check | Returns task array with model metadata |
| POST /api/v1/approvals/{id}/decision | Human approval decision | { decision: 'approved'\|'rejected', reason } |
| POST /api/v1/rag/ingest | Ingest & embed RAG document | { project_id, title, content } |

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| | | |
|---|---|---|
| POST /api/v1/rag/search | Vector semantic search with citations | { project_id, query, top_k } |
| GET /health | Readiness probe & cluster telemetry | Status, mode, strategy, DB health |

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# 8. Deployment & Operational Runbook

## Option A: Docker Compose Deployment (Recommended)

The platform includes multi-stage container definitions for full-stack deployment with Docker Compose:

```
# 1. Build and start complete stack in background
docker-compose up --build -d

# 2. Stream backend application logs
docker-compose logs -f backend

# 3. Teardown
docker-compose down
```

## Option B: Bare-Metal / Local Development

```
# Backend (Terminal 1)
.\venv\Scripts\Activate.ps1
python backend/run.py # Server: http://localhost:8000 (Docs: /docs)

# Frontend (Terminal 2)
cd frontend && npm run dev # Dashboard: http://localhost:5173
```

# 9. Backup, Disaster Recovery & Rollback Protocol

- **Automated MongoDB Atlas Backups:** Configure continuous point-in-time recovery (PITR) with daily cluster snapshots in the Atlas Console.
- **CLI Database Dump:** mongodump --uri="$MONGODB_URI" --out=/backups/$(date +%Y%m%d)
- **Container Rollback:** If a regression occurs during deployment, roll back immediately with docker-compose down && git checkout HEAD~1 && docker-compose up --build -d.
- **Production Checklist:** Ensure TLS termination (Port 443), restrict CORS_ORIGINS, and confirm Ollama models are pulled (ollama pull deepseek-coder:6.7b, ollama pull llama3.2:latest).

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| **Document Approved by:** | Ayush Srivastava |
| **Verification Standard:** | All 17 Pytest Suites Passed (100% Pass Rate) |
| **Distribution:** | Enterprise Engineering, Operations, and Auditing Teams |

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